



Nanomand: A Commercialization Strategy

March 13, 2026

Functional Nanomaterials On Demand

Microreactor-assisted Nanomaterial Deposition (MAND)

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Introduction

This document outlines our team's efforts in developing a commercialization plan for Nanomand, a materials technology startup based in Corvallis, Oregon. It includes an analysis of the client, focusing on the business overview, basic competitive comparisons, and key competitive advantages. The report also presents customer profile information, including industry experts and their insights, company pursuit targets, and interview recommendations. In addition, a market analysis examines the two potential industries to pursue, along with trends related to opportunity and demand. After that, an analysis of applications for Nanomand is presented, specifically looking at applications for Tooling, Injection Molding, and Semiconductor cooling applications. Finally, the document includes a commercialization analysis and recommendation covering the promotional strategy, product launch approach, and a discussion of the timing of the market entry.

Client Profile

Nanomand is a materials technology startup based in Corvallis, Oregon, and is currently part of the Advantage Accelerator program. The company is in the early stages of research and development, focused on commercializing patented microreactor-based nanomaterial synthesis technology developed at Oregon State University. One of Nanomand's primary advantages is its intellectual property. The company holds patents related to its material integration process, creating a barrier to entry for competitors attempting to replicate the technology. In addition, Nanomand benefits from a strong research foundation, with expertise in materials science and engineering supporting the development of its technology.

Nanomand's patented microreactor enables nanomaterial synthesis with a range of customizable properties for the consumer. This process also allows for the creation of defined microstructures using nanoparticles. As well as precise placement of nanoparticles, in regards to

volumes and locations within the final product. The reactor can be directly integrated into the manufacturing process, reducing supply chain complexity by eliminating the need to store and transport nanomaterials altogether. Nanomand's technology provides a strong foundation for exploring potential commercial applications.

Customer Profile

Our customer discovery process focused on identifying industries, companies, and experts that could help validate the commercial potential of Nanomand's technology. Nanomand's microreactor-assisted nanomaterial deposition process is designed to integrate directly into advanced manufacturing systems, our discovery efforts focused primarily on potential B2B customers where material performance and manufacturing efficiency are critical.

Nanomand's relationship with AddiTec, a company specializing in wire arc additive manufacturing (WAAM). AddiTec is currently trialing Nanomand's technology alongside its WAAM printheads to enable selective nanoparticle deposition during additive manufacturing. This integration could allow manufacturers to adjust material properties in real time during the printing process. Some examples of these adjustable properties include hardness, thermal resistance, and strength-to-weight ratio. The ability to tune these characteristics during production could provide manufacturers with more control over the final material properties than traditional additive manufacturing processes allow. From this relationship, our main takeaway is that it has the potential to extend additive manufacturing capabilities beyond conventional material limitations.

Based on both the technical capabilities of Nanomand's process and our research, we identified two industries that may have strong demand for this technology: the semiconductor industry and the tool coating industry, and a few more that we didn't do as much in depth research on.

The semiconductor industry can benefit from Nanomand's application. Semiconductor devices are extremely sensitive to heat and require materials with very precise thermal and structural properties. Nanomand's ability to integrate nanoparticles into materials for improved thermal dissipation and material property control could be valuable to semiconductor manufacturers. As computing hardware becomes more powerful, heat management continues to be a growing challenge, which increases the potential value of technologies that can improve thermal performance.

We believe that tool coating represents a viable market that may also be easier to enter. Tool coatings are used in machining and manufacturing processes to improve durability, reduce wear, and maintain performance under high temperatures. Important properties in this market include hardness, durability, and hot hardness, which refers to a material maintaining or increasing hardness as temperature rises. Nanomand's technology could potentially create coatings that provide improvements in these areas, which would be valuable in industries that rely heavily on machining and tooling performance.

We compiled a group of industry experts who could provide insight into additive manufacturing, machining, and advanced materials. However, for most experts we were unable to make direct contact because public contact information was limited. While we located several experts through LinkedIn, contacting them directly would have required a premium subscription. Instead, we identified individuals who could potentially provide valuable insight if contact were established in the future. Their full names are stated in the appendix of this document.

While most outreach attempts were unsuccessful due to limited contact, we were able to conduct an interview with Scott Campbell, whose background in machining and tooling made him one of the most valuable experts for evaluating Nanomand's potential in the tool-coating market. Campbell was intrigued by Nanomand's idea and believed the technology could create value in the tool coating space. This potential will be further explored later in this report.

Nanomand's technology is intended for industrial manufacturing processes, our customer discovery also included identifying potential early-adopter companies with the ability to evaluate and implement new materials technologies.

In the semiconductor industry, we identified Intel as a potential customer because of its large-scale semiconductor manufacturing operations and continued need for improvements in thermal management. We also identified Broadcom as another possible entry point into the semiconductor market.

For the tool coating and aerospace manufacturing markets, we identified Lockheed Martin as a potential early adopter due to its reliance on high-performance machining processes. Similarly, Boeing represents another potential entry point into the aerospace manufacturing industry, where improvements in tooling efficiency could have measurable production benefits.

Market Analysis

Research Efforts

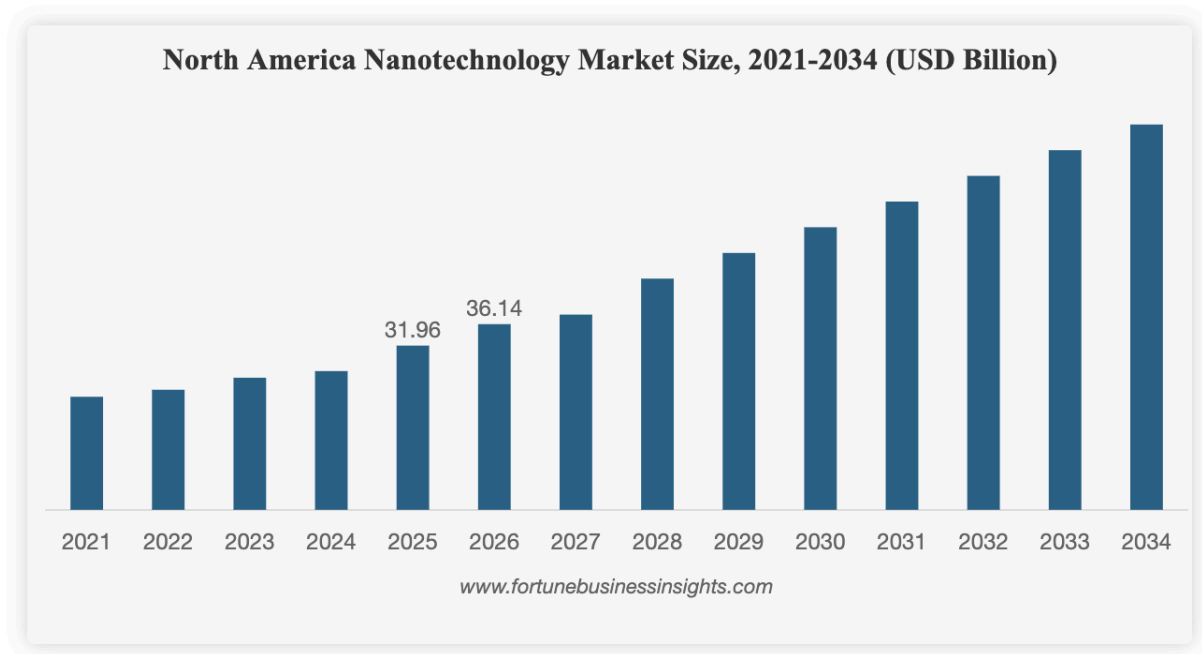
For our market research, our goal was to understand the size and direction of the broader nanotechnology market, then narrow that market to the areas that are actually relevant to Nanomand.

We began by looking at the overall global and North American nanotechnology market to understand its growth trajectory. From there, we filtered the broader market based on technical fit. Nanomand is a metal-on-metal material application system, so we excluded applications such as healthcare and other biologically focused nanoparticle uses, since those are outside the scope of what Nanomand's technology is built to address.

We also excluded nanoparticle applications tied to semiconductor silicon fabrication processes where Nanomand's current process would not apply. Instead, we focused on the areas where a metal-based nanoparticle integration system could reasonably create value.

We chose to emphasize the North American market because Nanomand is a U.S.-based company, so early commercialization in North America would likely be easier from both a business and regulatory standpoint.

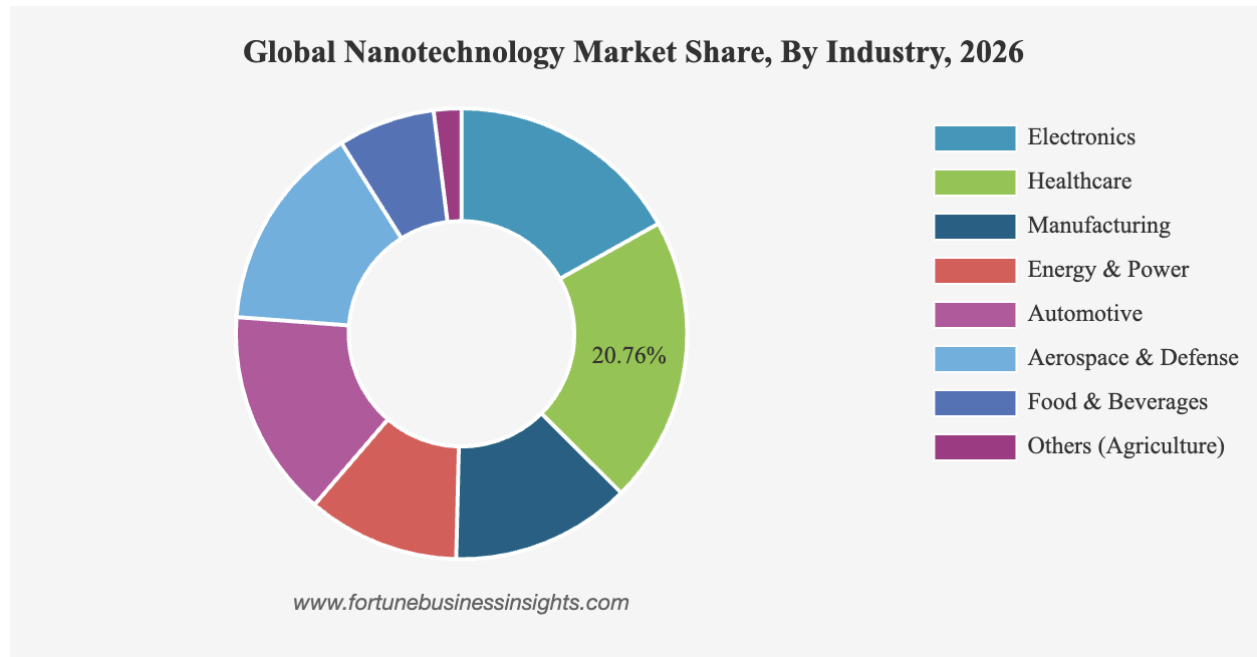
Within the figure 1 below, the North American market size's history and forecasted growth can be seen.



- Figure 1: A graph showcasing an estimated forecast of the North American nanotechnology market size (Fortune Business Insights, 2025).

At a high level, the nanotechnology market is both large and growing quickly. Fortune Business Insights documents the North American market was almost \$32 Billion in 2025, and that the global market at \$105 billion in 2025. It also documented that the global market size was projected to reach \$398 billion by 2034, with a 15.7% CAGR. We used this as a top-down signal that Nanomand is operating in a large and expanding macro market, not as a claim that the full market is directly available to the company.

After establishing the broader market size, we then looked at how nanotechnology demand is distributed across industries. A graph showcasing the distribution can be seen below in Figure 2.



- Figure 2: A chart showcasing the industry breakdown for the global Nanotechnology market share (Fortune Business Insights, 2025).

This chart shows that the largest industry segments include Healthcare at 20.76%, Electronics at roughly 18%, Automotive at roughly 15%, Aerospace and Defense at roughly 14%, Manufacturing at roughly 13%, and Energy and Power at roughly 11%. For Nanomand, the most relevant takeaway is not the largest segment overall, but rather the industries where metal-based material improvements could matter most. That is why we focused more heavily on electronics, manufacturing, aerospace and defense, automotive, and energy-related use cases, while excluding categories where Nanomand's process does not realistically apply.

Industry Trends

We then examined major market trends. Below, within Table 1, we've identified which ones appear most likely to translate into real commercial demand. In that table, we analyze the trend, the effect, and the opportunities from those trends that Nanomand could act on. We also give analysis on the real-world commercial demand of those trends.

Trends to Opportunities			Trends to Demand			
Description			"TAM"	"SAM"	"SOM"	
Mega Trend	Effects	Opportunities	Total Addressable Market	Serviceable Addressable Market	Serviceable Obtainable Market	Percent Annual Growth Rate
General AI Boom	There are many AI Data Centers being built due to companies competing for Compute power	Nanomand can help develop high powered cooling solutions	\$ 12,000,000,000	44%	8%	
				\$ 5,220,000,000	\$ 417,600,000	21.2%
High Efficiency Tools for cutting, drilling and milling	Increased demand for tools that can support efficiency/longevity of cutting, drilling, and mining efforts.	Nanomand can create high quality tool coatings that provide increased Life Span or Performance	\$ 5,150,000,000	65%	12%	
				\$ 3,347,500,000	\$ 401,700,000	5.2%

- Table 1: A table showing trends to opportunities along with their associated trends to demand information.

The first major trend is the general AI boom. Growth in AI infrastructure is driving rapid data center expansion, increasing the need for advanced thermal management solutions. This trend corresponds to an estimated \$12 billion total addressable market, with an estimated \$418 million serviceable obtainable market, and roughly 21.2% annual growth. This makes cooling a potentially high-upside opportunity for Nanomand, although the exact application has not yet been validated through direct expert input.

The second major trend is the rising demand for high-efficiency tools used in cutting, drilling, and milling. Companies in these industries continue to look for tools that last longer, reduce wear, and improve productivity. We estimate this area to house a \$5.15 billion total addressable market, with roughly \$402 million serviceable obtainable market, growing at 5.2% annually. Compared with cooling, this appears to be a more direct fit for Nanomand because performance in tooling is closely tied to measurable properties such as hardness, heat resistance, and durability.

Together, these two trends stood out because they combine a meaningful market size with plausible technical relevance for Nanomand. Cooling appears to offer the larger upside and faster growth rate, while tooling appears to offer a more straightforward near-term path for testing and application.

We also identified several additional trends within a report from Mordor Intelligence that may support future opportunities, even if they are not the clearest immediate focus. Below, in Table 2, we show an analysis of those trends, along with their estimated effects and the opportunities which Nanomand could leverage.

Trends to Opportunities		
Mega Trend	Effects	Opportunities
Battery anode breakthroughs using silicon nanowires for EV range	This can accelerate the shift towards more efficient EVs with improved battery performance, reduction in charging times, and increased longevity.	Nanomand can develop scalable materials that could boost EV range, charging performance, and overall longevity.
United States CHIPS Act support domestic nanosensor fabs	Semiconductor manufacturing will see increased demand as nanosensor Fabs (fabrication facilities) receive more government support, and thus additional funding.	Nanomand can leverage the increased government funding for nanosensor fabs.
Rapid adoption of quantum-dot displays by Asian TV OEMs	High adoption of nanometer-scale, semiconductor crystals will result in increased demand for mineral based nanoparticle application.	Nanomand could leverage the increased revenue and thus R&D expenditure from Asian TV OEM companies to insert themselves into their production line.

- Table 2: A table showing more trends to opportunities (Mordor Intelligence 2026).

For the first trend, battery anode breakthroughs using silicon nanowires showed an estimated +2.3% impact on the CAGR forecast, making them one of the stronger secondary growth trends we identified. Rapid adoption of quantum-dot displays showed roughly +1.6%, while CHIPS Act support for domestic nanosensor fabrication showed roughly +1.4% (2026). These trends help show that nanomaterials continue to receive investment and attention across batteries, semiconductors, and advanced displays, even if not all of these areas are immediately actionable for Nanomand's current metal-on-metal process.

To summarize the market opportunity, we used a TAM, SAM, and SOM framework. In this report, TAM refers to the full size of the broader market, SAM refers to the portion of that market where Nanomand's technology appears technically applicable, and SOM refers to the smaller share Nanomand could plausibly reach over the next several years, depending on the business model and market path they choose. This summary can be seen below in Table 3.

"TAM"	"SAM"	"SOM"	Annual % Growth
\$ 105 B	\$ 32 B	\$ 3.5 B	15.7%

- Table 3: A table showcasing market data for the semiconductor industry, as well as how much is obtainable for Nanomand.

Overall, our market research suggests that Nanomand is operating in a fast-growing macro market, but that only a subset of that market is truly relevant to its current technology platform. Within that subset, cooling-related applications and high-efficiency tooling appear to be the strongest current opportunity areas. Cooling offers the larger growth opportunity, while tooling appears easier to connect to measurable performance improvements and earlier real-world testing.

Competitor Research

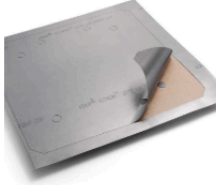
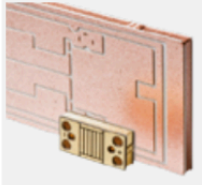
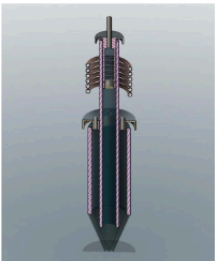
To understand Nanomand’s potential position within the semiconductor cooling market, we focused our competitor analysis specifically on heat-spreading materials. Heat spreaders are a critical component of chip cooling systems, responsible for distributing heat away from localized hotspots on semiconductor devices and transferring it to a cooling system, such as a heatsink or a liquid-cooling interface.

We chose to focus on heat-spreading mechanisms because this is the area where the most reliable performance data were available for comparison. While Nanomand’s technology may improve several material properties, thermal conductivity is currently the most measurable metric that can be directly compared to existing solutions.

The semiconductor thermal management market is currently dominated by traditional metals, such as copper and aluminum, which together account for the majority of deployed heat-spreading materials. Copper accounts for roughly half of the market due to its relatively high thermal conductivity and manufacturability, while aluminum provides a lightweight and low-cost alternative used in many consumer and industrial applications.

However, as semiconductor power densities increase, particularly in high-performance computing, traditional materials are approaching the limits of their cooling performance. This has led to the development of advanced materials such as graphite sheets, graphene-based spreaders, and engineered thermal interface materials designed to improve heat transport.

The competitor comparison as seen below in Table 4 describes the results of our research into these advanced materials and the industry leaders that provide them.

Company Name	Your Client’s Company Name	Primary Competitor’s Name	Competitor’s Name	Competitor’s Name
Product Name	Nanomand	Rogers Corporation’s Curamik cooling solutions	Neograp’s eGraf SpreaderShield Heat Spreaders	T Global Technologies’ T68 Synthetic Graphite Sheet
Unit Price (per kg)	\$14 (copper based)	\$14 (copper based)	\$13	\$6
Thermal Conductivity (W/mK)	700 (through-plane)	400 (through-plane)	350-600 (in-plane) 3 (through-plane)	1500 (in-plane) 5 (through-plane)
				

- Table 4: A table documenting Nanomand and their relative competition, as well as various comparable attributes between the companies’ competing products.

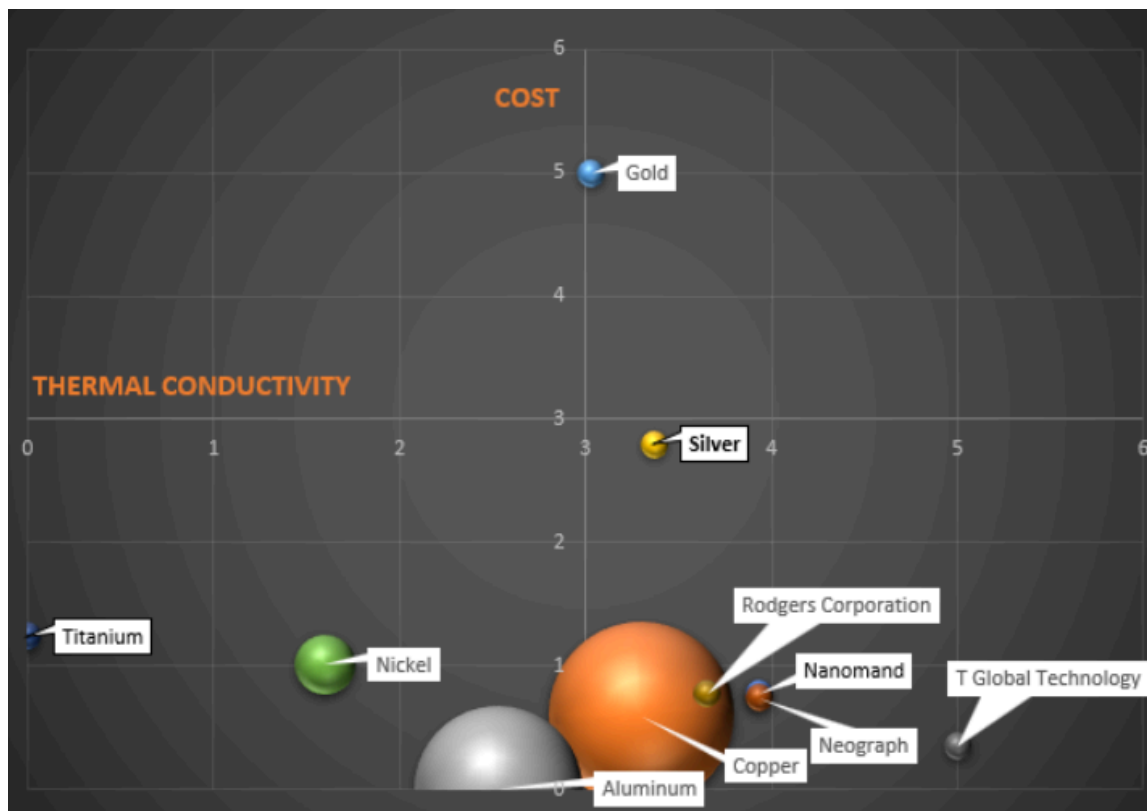
When comparing these materials to Nanomand's technology, two key dimensions become important: thermal conductivity performance and economic feasibility. While Nanomand's costs are roughly equivalent to other copper-based heat spreaders, we can see that Nanomand has a much higher through-plane thermal conductivity, and while the cross-plan thermal conductivity is lower than some other specialized graphite/graphene based materials, it is our perspective that Nanomand has enough of a through-plane performance increase to be competitive in the heat spreading market.

Cross-Plane Thermal Conductivity Analysis

The competitor analysis chart seen in Figure 3 below compares several candidate materials across three variables:

- Thermal conductivity (x-axis)
- Cost per kilogram (y-axis)
- Current market share (bubble size)

To enable comparison across materials with dramatically different cost and conductivity values, the chart uses logarithmic scaling for both cost and thermal conductivity.



- Figure 3: A chart showing the relative market size, cost, and cross-plane thermal conductivity of Nanomand compared to competing companies and materials.

Traditional materials such as copper and aluminum appear in the lower-cost region of the chart and have the largest bubble sizes because of their widespread adoption. These materials remain dominant primarily because they are inexpensive, easy to manufacture, and compatible with existing production infrastructure.

More advanced materials, including graphite and graphene-based heat spreaders, occupy a different region of the chart. These materials exhibit extremely high in-plane thermal conductivity, allowing heat to spread efficiently across the material's surface. However, their through-plane conductivity is relatively low, making them less effective at transferring heat vertically through the material to a heatsink or cooling system.

Through Plane Thermal Conductivity Analysis

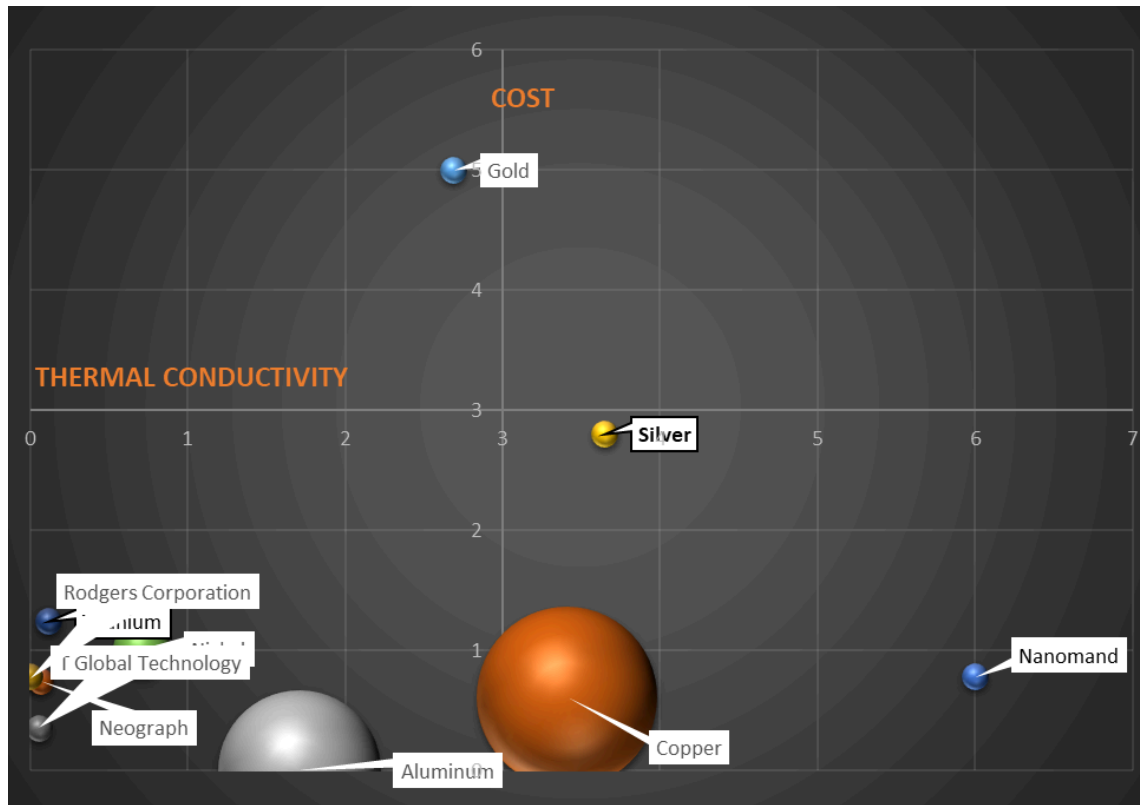
The distinction between in-plane and through-plane thermal conductivity is an important factor in evaluating thermal management performance. Materials such as graphite excel at spreading heat laterally but are less effective at removing heat from the chip stack. As a result, they are often used as part of layered thermal solutions rather than as standalone heat spreaders.

Nanomand's technology presents a potentially unique position within this competitive landscape. Based on the currently available data, Nanomand materials demonstrate relatively high thermal conductivity in both the in-plane and through-plane directions. This balanced thermal transport behavior distinguishes it from many existing advanced heat spreader materials that primarily excel in only one direction of heat transfer.

The next competitor analysis chart seen below in Figure 4 compares several candidate materials once again across these three variables:

- Thermal conductivity (x-axis)
- Cost per kilogram (y-axis)
- Current market share (bubble size)

Notably, this chart showcases the through-plane thermal conductivity of those same competitors and materials.



- Figure 4: A chart showing the relative market size, cost, and through-plane thermal conductivity of Nanomand compared to competing companies and materials.

When focusing specifically on **through-plane thermal conductivity**, Nanomand appears to outperform several alternative advanced materials currently used in the market. This suggests that Nanomand may be particularly effective at moving heat away from localized hotspots and transferring it into external cooling systems.

From a cost perspective, Nanomand materials appear to fall within a similar price band as other advanced thermal management materials currently used in high-performance semiconductor applications. While these materials are more expensive than traditional copper or aluminum, they are often justified in applications where improved thermal performance enables higher chip performance or improved reliability.

However, it is important to note that while Nanomand shows promising thermal conductivity characteristics, the real-world impact of these improvements on system-level cooling performance remains uncertain. Semiconductor cooling systems involve multiple layers of materials, thermal interface layers, and mechanical interfaces that influence overall heat transfer.

Because of this complexity, the advantage observed in material-level thermal conductivity does not automatically translate into proportional improvements in chip cooling performance. Further modeling and experimental testing would be required to determine the magnitude of the benefit within real computing systems.

Despite this uncertainty, the current analysis suggests that Nanomand may occupy a unique position among advanced heat spreader materials by combining relatively strong in-plane and through-plane thermal conductivities at a cost comparable to other high-performance thermal materials.

For this reason, while the semiconductor cooling market presents higher barriers to entry than the tooling industry, the performance characteristics observed in the current data suggest this application area warrants further investigation. Additional modeling and experimental validation would help determine whether Nanomand's materials provide meaningful advantages in real-world thermal management systems used in high-performance computing environments.

Potential Industrial Applications

Tooling Industry Applications:

Relative to the semiconductor cooling market, the tooling and advanced manufacturing sector represents a more accessible and practical entry point for Nanomand's technology. Tooling performance is heavily dependent on material properties, and improvements in hardness, wear resistance, friction characteristics, and thermal conductivity directly translate into measurable improvements in manufacturing performance.

Modern cutting tools rely primarily on carbide substrates with thin-film coatings such as TiN, TiAlN, or AlTiN applied through physical or chemical vapor deposition. These coatings are optimized for wear resistance and thermal stability during high-speed machining operations. However, the performance ceiling of these coatings is constrained by the underlying material system and coating architectures currently available.

Nanomand's nanoparticle integration process presents an opportunity to introduce new material behavior into these systems. If the technology can improve hardness, thermal conductivity, or wear resistance beyond the current coating standards, the resulting performance gains could translate into longer tool life, reduced cutting forces, and improved machining efficiency. At the current stage, the magnitude of these improvements is still unknown and must be determined experimentally. However, the tooling industry offers a controlled environment where improvements can be validated quickly and quantitatively.

A practical development pathway would involve applying Nanomand's coating process to commercially available carbide or high-speed steel cutting inserts. These inserts are inexpensive, standardized, and widely used in CNC machining operations, making them ideal for comparative testing.

Once coated, these tools could be evaluated using standard machining tests on CNC equipment. Performance metrics such as spindle power consumption, cutting forces, tool wear rate, and heat generation can all be measured directly during machining operations. These metrics would allow the Nanomand team to determine whether the coating provides measurable advantages relative to existing PVD or CVD coatings.

If improvements are demonstrated, commercialization could occur through several pathways. Nanomand could license the coating technology to existing tool manufacturers, partner with tooling companies to co-develop enhanced tools, or produce specialty cutting tools directly. Because cutting inserts are already distributed through established industrial marketplaces and distributors, market entry could be achieved without requiring major changes to purchasing workflows.

Injection Molding Applications:

Injection molding represents another high-potential opportunity within the tooling sector. The process is used extensively across manufacturing industries, producing plastic components for automotive, electronics, consumer goods, and medical devices.

Injection molds are typically manufactured from hardened tool steels designed to withstand extremely high clamping forces and repeated thermal cycling. Mold performance depends on maintaining dimensional stability while effectively removing heat from the molten plastic during each cycle.

Cooling is a dominant factor in injection molding cycle time. In many molding processes, the cooling phase accounts for a majority of the total production cycle. Parts must cool sufficiently before ejection to avoid deformation, shrinkage inconsistencies, or surface defects. Because of this, mold designers often prioritize efficient heat transfer when designing mold cavities and cooling channels. Materials with higher thermal conductivity can reduce the time required to remove heat from the molded part, allowing the mold to cycle more quickly.

Nanomand's technology may provide a meaningful advantage if the nanoparticle integration process can increase the thermal conductivity of mold materials without compromising hardness or structural stability. Even small improvements in heat transfer could reduce cooling times, leading to measurable increases in production throughput.

In high-volume manufacturing environments, small improvements in cycle time can generate large economic benefits over the lifetime of a mold. Because molds are often used to produce hundreds of thousands or millions of parts, a small percentage reduction in cycle time can significantly increase production capacity. For this reason, injection molding tooling represents a promising application area if Nanomand's material improvements can simultaneously maintain hardness and improve thermal conductivity.

Semiconductor Cooling Applications:

The semiconductor cooling industry represents another potential market for Nanomand's materials due to the increasing thermal management challenges associated with modern computing hardware. High-performance computing systems, particularly those used for artificial intelligence workloads, generate extremely high heat densities. Effective thermal management is critical for maintaining performance and preventing thermal throttling in these systems. Nanomand's technology may provide value in this space through materials with extremely high thermal conductivity. If integrated into heat spreaders, heat sinks, or thermal interface materials, these materials could potentially improve heat transfer away from semiconductor devices.

However, entry into the semiconductor industry presents several significant challenges. Semiconductor manufacturers require extensive validation and reliability testing before adopting new materials in production environments. Qualification timelines can be long, and the industry is dominated by large established companies with significant research budgets.

Additionally, several alternative cooling technologies are currently being explored. One notable example is microfluidic liquid cooling, in which coolant flows through microscopic channels embedded within or near semiconductor devices. These systems can provide highly efficient heat removal and are currently being investigated by major technology firms for next-generation AI hardware.

Another technical consideration is the distinction between cross-plane and through-plane thermal conductivity. Depending on the geometry of the heat source and the cooling system, improvements in one direction of heat transfer may be more valuable than improvements in another.

Because of these uncertainties, additional modeling and experimental testing will be required to determine whether Nanomand's materials provide meaningful advantages in real semiconductor cooling systems.

Experimental Validation Strategy:

Because the exact magnitude of Nanomand's material performance improvements remains uncertain, the most critical next step for commercialization is experimental validation. Several relatively simple experiments could provide valuable insight into the technology's potential applications:

- Machining experiments using Nanomand-coated carbide inserts
- Thermal conductivity measurements on nanoparticle-enhanced metals
- Injection molding cooling experiments using modified mold inserts

These experiments would allow the team to quantify improvements in hardness, wear resistance, and thermal conductivity relative to existing materials.

Once measurable performance improvements have been demonstrated, the Nanomand team will be in a much stronger position to evaluate which industry applications provide the greatest economic opportunity.

Commercial Analysis and Recommendations

Commercialization Preface

There are currently two promising industries Nanomand could commercialize into with their specific ability to improve material properties and application process, depending on the returns expected and the risks they are willing to take. One is the semiconductor industry. The biggest advantage of this industry is the market size and growth driven by AI chips and data centers. Higher performance chips could be generated via improved heat transfer materials, something Nanomand's materials could provide over traditional, established heat spreaders. This advantage over traditional heat spreaders could be enough to overcome the engineering process inertia of larger companies, if a high enough difference in performance is demonstrable by Nanomand. However, this industry has very high barriers to entry. Often requiring strong partnerships with chip makers, along with the ability to quickly fulfil bulk orders for parts, and needing parts to be able to last years with as minimal maintenance as possible. Almost all of the parts that Nanomand would be integrated into would be considered large capital investments where the companies purchasing want a degree of certainty that their investment will work.

The next is the tool coating industry. Tool coatings would provide a more accessible entry point for Nanomand. Cutting tools are small and relatively easy to prototype and test, and the industry often adopts improvements faster because tooling is frequently replaced. With tools being a consumable that is not a capital expenditure, companies are more flexible in trying a variety of tools, ultimately resulting in a lower average engineering process inertia. The challenge then is that the market is highly competitive due to the ease of switching products. Nanomand would need to show clear performance gains over competitors to stand out and capture market share. Additionally, in a competitive market, design improvements occur more quickly, requiring quicker iterations and much more frequent improvements from Nanomand than in the Semiconductor industry. Furthermore, the tool coating industry is not as big as the semiconductor industry and would provide lower returns after market entry and the establishment of Nanomand's presence.

Ultimately, both industries present strong opportunities while representing different return-to-risk ratios. Semiconductors offer larger long-term potential, while tool coatings may allow faster testing and earlier commercialization. The decision will likely come down to the resources available to Nanomand, how quickly they want to move toward market entry, and what degree of risk is acceptable to them. Regardless of which industry is selected, as both deal with B2B, our commercialization strategy remains largely the same between the two.

Advertising & Public Relations Strategies and Platforms

Our recommended strategy for advertising and public relations focuses primarily on targeted outreach toward engineering professionals in their targeted industries. This would focus around a personal selling approach with efforts directed towards creating and maintaining connections

with companies. To that end, it's important to target the decision making individuals of the target company. The primary method for generating connections would be through industry conferences and trade shows (*See Appendix for Recommended Shows*). Attending and connecting with individuals at these events will generate the most high-quality leads with the highest likelihood of reaching sales conversion.

Additionally, It's important for Nanomand to build up a bigger presence in the targeted industry, alongside the trade shows and conferences. Complementary promotional activities would include:

- 1) Appearing within engineering newsletters
- 2) Warm introductions from connections
- 3) Cold outreach

In line with the theme of targeted outreach toward professionals, Nanomand's complementary promotional activities would be highly focused around a personal selling strategy. Alongside the standard brand building activity of advertising in newsletters, Nanomand would generate leads from reaching out to key individuals within companies where possible to generate new connections from nothing, while simultaneously probing existing connections for new opportunities.

Product Launch

We recommend Nanomand conduct their business as an advanced coating supplier, while simultaneously conducting custom material R&D for companies. This pathway would allow for more applications to be developed through partnerships with companies, and see where the demand is for the product. If there is significant demand for a specific application or material recipes built, then it would be relatively simple to move into licensing or Productizing depending on the structure of partnerships. Additionally, this method allows for Nanomand to not only generate their own learning effects, but to also take advantage of the learning effects of their partner companies as they develop the coating within the collaboration.

For their Product Launch, we recommend that Nanomand focus on initial, smaller scale trials, collaborating closely with specific target companies in their chosen industry. Working hand-in-hand with their specific partner companies to develop custom tailored products to the companies' specific specifications will let Nanomand generate both learnings for how Nanomand works in a real production environment, and also a track record that they can use to generate additional leads to work with more companies and build their brand.

Beachhead Strategies:

To secure their position in their desired market and establish their presence, we recommend a few activities for Nanomand. Firstly, it is necessary to enter into trial collaborations with target companies. After building a connection with the company through the previously listed promotional methods, entering into a collaboration with them and fulfilling an order for them would be the best way to build a brand directly within the selected industry. This would be the

most effective approach and would have Nanomand working with a small number of companies to develop real-world applications for Nanomand. These trials provide a great way to generate test data and build case studies under real-world conditions that demonstrate Nanomand's performance, such as tool wear rates or thermal performance. These trial projects would not only provide Nanomand with solid data, but could also lead to additional leads at associated or adjacent companies that may have a need for Nanomand. Additionally, they would help in the process of building professional networks by providing credibility to Nanomand from having worked with them.

Another method we recommend for establishing themselves in a market is to collaborate with universities and industry groups. Working with other engineering labs at Oregon State University or other related universities can help provide a level of technical validation to Nanomand's technology, and generate peer-reviewed research describing its performance increases. Both the trial runs and research lab collaborations provide practical pathways for Nanomand to enter the market and demonstrate real world performance.

Timing of Entry

Each specific industry we've recommended for Nanomand to pursue is at a different stage in its lifecycle, and requires some very different considerations based on these qualities. Looking at the semiconductor industry, it is currently mature but is undergoing rapid changes and development due to the pressing demands placed on it via the advancement of AI and of the large-scale development of data centers. Because this industry is mature, Nanomand has a high barrier to entry as they need to overcome the inertia present within existing manufacturing processes. Additionally, there are qualification and reliability testing requirements that Nanomand would need to satisfy for their product, which contributes to the long product adoption cycles. This is why Nanomand will need to conduct much of their initial promotional efforts utilizing personal selling methods, making sure to build relationships and presence as a real option that provides significant value over competitors.

For the Tool Coating industry, it's a mature manufacturing-support industry that is characterized by steady incremental growth and innovation and high competition. While this industry is mature and has high levels of competition, it also has less inertia within business' production process, faster testing cycles, and lower product adoption periods. This industry is characterized by the high levels of competition and product switching upon the release of better products. Manufacturers are able to quickly adopt new improvements in tool coating technology into their production pipelines. Additionally, as the adoption cycles are shorter, Nanomand could generate very early validation from real businesses, as well as early revenue to help fund future efforts. While this industry would allow for a quicker shift into productizing if Nanomand wished to pivot that way, we still recommend entering as an advanced coating supplier that also does custom material R&D. Early promotional efforts and brand building in this industry would thus also focus mainly around personal selling, as the specifics of how Nanomand would integrate into the production pipelines of manufacturers would need to be discussed in a detailed back and forth between the two parties. This industry would be easier for Nanomand to enter and

could provide some faster results, but only if Nanomand can provide clear performance advantages to the customers.

Commercialization Considerations:

At the current stage, the economic viability of specific applications will depend on both performance improvements and coating cost. Preliminary estimates suggest that a coating cost between approximately \$2 and \$10 per component could be competitive for many industrial tooling applications. However, acceptable pricing will ultimately depend on the value created through increased tool life, improved machining performance, or reduced production cycle times.

Because tooling applications allow rapid testing and clear performance measurement, this sector represents a solid starting point for commercialization. Demonstrating measurable improvements in tool performance would provide strong validation of the technology and could open the door to broader industrial applications.

Summary

- **Overview of Nanomand and Technology**
 - Corvallis based startup commercializing patented microreactor nanomaterial synthesis technology
 - Enables precise nanoparticle placement and customizable material properties
 - Integrates directly into manufacturing, reducing supply chain complexity
 - Strong intellectual property and materials science expertise provide competitive advantage
- **Customer Discovery and Target Markets**
 - Focus on B2B industries where advanced materials improve manufacturing performance
 - Early collaboration with AddiTec to integrate nanoparticle deposition into additive manufacturing
 - Potential customers as follows: Intel, Broadcom, Lockheed Martin, and Boeing
 - Expert interview with Scott Campell supported potential value in tool coating applications
- **Market Opportunities**
 - Two primary industries identified: semiconductors and tool coatings
 - Semiconductor: strong demand for improved thermal management in AI chips and data centers
 - Tool coatings: potential improvements in hardness, durability, and heat resistance for machining tools
 - Tooling market offers faster testing and adoption compared to the highly regulated semiconductor industry
- **Competitive and Technical Position**
 - Semiconductor cooling market currently dominated by copper and aluminium heat spreaders

- Emerging competitors include graphite and graphene based materials
- Nanomand materials may offer balanced in-plane and through-plane thermal conductivity
- Performance advantages appear promising but require additional modeling and real world testing
- **Commercialization Strategy**
 - Focus on targeted B2B marketing through industry conferences, trade shows, and direct outreach
 - Emphasize personal selling and relationship building with engineering decision makers
 - Begin with collaborative pilot trials and custom material development for partner companies
 - Use early projects to generate performance data, case studies, and industry credibility
- **Market Entry Recommendation**
 - Tool coating industry recommended as a practical initial entry point due to faster testing and lower barriers
 - Semiconductor market offers larger long term returns but requires extensive validation and partnerships
 - Early experiments and pilot trials will be essential to quantify performance improvements
 - Successful demonstrations can lead to licensing, partnerships, or specialized product development

Conclusion

Nanomand's patented microreactor technology provides a unique capability to engineer materials with precise nanoparticle placement, creating performance advantages in durability, thermal conductivity, and hardness. Customer discovery and expert interviews indicate that these advantages are relevant to high-performance applications, particularly in the semiconductor and tool coating industries.

Market analysis highlights two potential entry points: the semiconductor industry, which offers a large, high-growth market but with higher adoption barriers, and the tool coating industry, which allows for faster prototyping and market entry, particularly within aerospace applications. Competitor analysis shows that Nanomand can deliver superior material performance while maintaining a competitive cost position in specialized applications.

Based on these findings, we recommend that Nanomand evaluate the resources and timing required to pursue either industry. A focused entry strategy, supported by strategic partnerships, targeted promotional efforts, and technical collaboration, will allow the company to validate its technology in the market. With demonstrable value to early adopters, and position itself for broader commercialization over time.

References

Fortune Business Insights. (2025). *Nanobiotechnology Market Size, Share, and Industry Analysis*.
<https://www.fortunebusinessinsights.com/nanotechnology-market-108466>

Mordor Intelligence. (2026, January 20). *Nanotechnology Market Size & Share Analysis - Growth Trends And Forecast (2026 - 2031)*.
<https://www.mordorintelligence.com/industry-reports/nanotechnology-market>

Appendix

Interview Questions

From our research, expert interview, and conversations with Nanomand, we developed a set of core questions that need to be answered in order to further validate the product. These questions are intended to clarify the value proposition and identify the industries where adoption would be most likely.

Key questions include:

- Do potential customers find value in the concept of Nanomand?
- What specific problem does this material solve that currently costs companies money?
- What measurable performance improvement could Nanomand deliver, and how significant would it be?
- Who within these organizations would have the authority to adopt or test this technology?
- What is the fastest way for a customer to test or pilot this technology within their current manufacturing process?
- Which industry could adopt this technology without long certification or regulatory delays?

Expert Contacts

Amy M. Elliott at Oak Ridge National Laboratory
Chinedum Okwudire at Ulendo, a 3D printing company.
Travis Mayberry at Divergent
Richard Wirz at Boeing & Oregon State University
Bahman Abbasi at Oregon State University
Benjamin Adam at Oregon State University
Bob Kaplan, Oregon State University Alumnus
Scott Campbell, Oregon State University

Potential Customer Contacts

Intel: Marcos Hernandez (Module Engineer) - Semiconductor Industry
Broadcom: Derui Kong (Principle Design Engineer) - Semiconductor Industry
Lockheed Martin: Nathan Richardson (Associate Fellow) - Tool Coating (Aerospace Focus)
Boeing: Will Newberry (Process Engineer) - Tool Coating (Aerospace Focus)
Silfab Solar Inc.: Suyash Karingwar (Product Engineer) Solar Industry
Lucid Motors: Shantanu Kulkari (Senior Manufacturing Engineer) Electric Vehicle Industry
BWX Technologies: Jeffrey Bernstein (Manufacturing Engineer) Nuclear Industry

Advanced Materials Show

- Tool Coating Industry Related
- October 6-7, 2026
- Pittsburgh, Pennsylvania

SEMCON West

- Semiconductor Industry Related
- October 13-15, 2026
- San Francisco, California

Links to Resources on Injection Molding, Tooling and Semi Conductors

Carbide Cutting Tool coated in Titanium: <https://www.mscdirect.com/product/details/62287933>

What Tool to use in machining:

<https://www.palbit.pt/en/material-properties-and-tool-selection>

Machineability: <https://en.wikipedia.org/wiki/Machinability>

Explainer on injection molding cooling <https://www.youtube.com/watch?v=TOcJsxypaRw>

Injection molding cooling article:

<https://rjginc.com/injection-molding-cooling-time-a-breakdown/>

Injection molding cooling article:

<https://www.osfplastic.com/en/blog/understanding-manufacturing-importance-of-cooling-in-the-injection-molding-process>

Cooling in AI and a potential competitor:

<https://news.microsoft.com/source/features/innovation/microfluidics-liquid-cooling-ai-chips/>